

AKASH KARMAKAR

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Education

Baruch College, Zicklin School of Business, MS in Finance-Fintech - New York Aug 2024 – May 2026

- GPA: 3.6/4.0 , **Coursework:** Quantitative Investments, Programming, Machine Learning in Investment

Jadavpur University, Bachelor in Engineering - Kolkata, India July 2018 – July 2022

- GPA: 3.7/4.0 , **Coursework:** Machine Learning, Mathematics 1 & 11, Numerical Methods

Experience

AIMPAAS, Alternate Investment Management Platform as a Service , Software Programmer Intern – New York Oct 2025 – Jan 2026

- Working with multi-database infrastructure (MariaDB, MongoDB, Redis) supporting hedge funds and top banks, enhancing system reliability.

WorldQuant, Quantitative Research Consultant – New York April 2025 – Sep 2025

- Designed and simulated trading signals across US, EMEA and APAC markets, achieving exceptional performance metrics, including a portfolio Sharpe ratio of 2.24 and a RoMaD of 9.21 in simulation.
- Optimized market-neutral portfolios and created alphas delivering annualized returns exceeding 15% with low drawdowns, validated through multi-regime analysis, parameter sensitivity tests and robustness checks to ensure stability and minimize overfitting.
- (National Finalist) US rank 3 and global rank 14 out of 76,867 in the International Quant Championship 2025.

FBN Securities, Developer Intern (Fall 2025) – New York Sep 2025 – Dec 2025

- Developed software to convert video data into transcripts using speech-to-text pipelines and applied NLP and AI techniques for sentiment analysis to classify commentary related to specific stocks, enabling traders to extract actionable insights from unstructured content and improve decision-making efficiency.

Goprac, Software Engineer – Bengaluru, India June 2023 – April 2024

- Automated the interview question upload flow with auto-validation and cluster balancing, making uploads faster, safer and more accurate while cutting labor time and cost.
- Collaborated with the product team and led the integration of an AI-driven personalized feedback system using skills-based scoring, optimizing assessment workflows, reducing friction by removing manual review and improving efficiency for 36,000+ users.

Celebal Technologies, Data Scientist – Jaipur, India July 2022 – June 2023

- Designed prompts using Chain-of-Thought for Text2SQL, Multi-intent classification and medical report analysis.
- Assessed advanced data mining techniques and conducted exploratory data analysis (EDA) on unstructured data, leading to significant insights for feature engineering.

Technical Skills

Languages/Frameworks: C , C++ , Python , Java , Javascript , Node.js , React.js , SQL , PHP , Rust , Tensorflow

Technologies: Postman , Linux , NLP , AWS , MongoDB , SQL Server , XCode , Redis , Power BI , Excel , OpenAI

Achievements

- (National Finalist) US rank 3 and global rank 14 out of 76,867 in the International Quant Championship 2025.
- Secured 3rd place in the Zicklin Graduate Case Competition 2025.
- Ranked globally 1056th in Google Kickstart 2021 coding competition round E.

Projects

Dynamic Sector Rotation with Machine Learning Github link

- Designed and implemented a LightGBM-based sector rotation strategy integrating PCA and K-Means clustering for market regime detection and ETF allocation optimization, achieving a Sharpe ratio of 1.12 in-sample. Conducted robust backtesting across 2014–2025 market regimes, outperforming the Pairs Portfolio benchmark in out-of-sample returns (8.76% vs. 7.35%) while controlling drawdowns.